(Please write your Enrollment Number) Enrollment No. 11001172022

MID-TERM EXAMINATION

Course Name : B.Tech. CSE AI Semester : 5

(October 2024) OFF LINE mode

Subject Code: BAI-301	Subject: Machine Learning
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Describe the purpose of validation data in model training.		CO1
	b) What is feature extraction, and why is it important in machine learning?		CO1
	c) Define Ridge Regression and explain how it addresses the issue of multicollinearity in linear regression.		CO2
	d) Explain the basic concept of Support Vector Machines (SVM). What is the significance of the margin in SVM?		CO2
	UNIT I		CO Mapping
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) What is K-Fold Cross Validation? Given a dataset containing 10K samples apply: (i) 5-Fold Cross Validation (ii) Stratified 4-Fold Cross Validation Calculate how many samples will be in the training and testing sets for each fold.		CO1
	b) Explain the importance of data cleaning and integration in machine learning. What are the common techniques used for handling missing or inconsistent data?		CO1
	c) A binary classification model is evaluated on a test set of 150 samples. The confusion matrix obtained from the model's predictions is as follows:		CO1

	Predicted Positive	Predicted Negative
Actual Positive	70	10
Actual Negative	5	65

(i) Construct the confusion matrix and label its components (True Positives, True Negatives, False Positives, and False Negatives).

(ii) Calculate the following metrics for the model:

	• Accuracy • Precision • Recall • F1 Score		
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Define Least Mean Square Regression and explain the process of finding the optimal coefficients.		CO2
	b) Explain the Naïve Bayes Classifier and its underlying assumption of feature independence. Describe its application in text classification and discuss its advantages and limitations.		CO2
	c) Explain the logistic regression model and its application in binary classification.		CO2